RESULTS: PKM fall-degree study, the t = 12 bracket (jobs pfd-x-t12, Amendment 7 B)

Reading (applied by hand from the rule text; the Amendment 7 D scorer patch is not committed): at Kummer m = 2, P1, t = 12, the bracket gives d_F = 6 on both draws. The median is 6 and c*(12) = 6. Amendment 7 D.5 reads **"holds at t = 12"**. [pcc:unverified]

- Cap 5 does not refute on either draw (so d_F >= 6). Cap 6 refutes on either draw (so d_F <= 6).
- Both draws have grid_zeros = 0: neither is decomposable.
- The cell predicts the same value as t = 11, so it cannot tell "C-step" from "flat at 6". t = 13 and t = 14 stay undecided (see "What this does not show").

Every number below is [pcc:unverified]: no pcc claim id exists for any of them.

Evidence classes: **measured** = read from the rows under results/X1..X4/; **derived** = a comparison or the floor in c* applied to those rows by the scratchpad script of the archive in section 6.

1. What was run (measured)

Four command jobs on Iso Lab EKS Spot, one capped F4 run each (f4_fp_tower, cap:12:<draw>:<cap>:12800000000), all on the frozen image and binaries of pfd-a. The cell is Kummer m = 2, p = 2013265921 (P1), t = 12 (N = 24), random targets, draws 0 and 1 (target_index), --budget 86400, --max-dense 12800000000.

job	card (launch)	draw	cap	request file	protocol_sha256 (request)
pfd-x-t12-k0-c5-20261108 (X1)	X1-1	0	5	requests_r2f/pfd-x-t12-k0-c5.json	cc6573af0782aa2e...
pfd-x-t12-k1-c5-20261108 (X2)	X2-1	1	5	requests_r2g/pfd-x-t12-k1-c5.json	bbd17530ee06205a...
pfd-x-t12-k0-c6-20261108 (X3)	X3-1	0	6	requests_r2g/pfd-x-t12-k0-c6.json	bbd17530ee06205a...
pfd-x-t12-k1-c6-20261108 (X4)	X4-1	1	6	requests_r2g/pfd-x-t12-k1-c6.json	bbd17530ee06205a...

- **Two protocol hashes.** X1 was launched from requests_r2f/ (hash cc6573af...). X2 to X4 were launched from requests_r2g/ (hash bbd17530...), after Amendment 13 (R2-B) added the m = 4 null cells. Per its provenance note, Amendment 13 changes no field of any other request except provenance. Each receipts/X*/launch/request.json equals the request file named in the table (compared, derived). For each cell, the r2f and r2g request differ only in provenance.protocol_sha256 and provenance.note; the job block (command, image, cpu, memory, seconds) is identical (compared, derived).
- **Resources per request:** cpu 12, memory 56Gi, seconds 90000, isolation_profile standard, retries 0, no instance pin (Amendment 12, price guard). host.txt of every pod: nproc 12, rayon: 12.
- **Image:** 104186811093.dkr.ecr.us-east-2.amazonaws.com/isolab/tasks@sha256:0d6b2ecd6df3f5f092fe4b6179986b041eb7d6be3b94e52d92809bdab2ff3ae8. binary.sha256 (each pod): source revision f13a690df016d0a457bd40579fbb9d054f486adf, pkm_tower_pilot f78f3959..., prime_fall_degree_tools 7e8a6703..., pfd_grid_zeros 962833611....
- **Node type:** not recorded. host.txt gives Intel Xeon Platinum 8488C, MemTotal about 61.8 GiB (64776596 to 64776600 kB), kernel 6.18.54-121.166.amzn2023.x86_64. Any instance type named from that is an inference.
- **Pods and dates (UTC):** X1 pod 9f2460de... 2026-11-10 00:27:37 to 00:27:48; X2 pod 16cf635f... 01:25:40 to 01:25:52; X3 pod b6596acd... 01:34:20 to 02:04:44; X4 pod 6cb84cac... 02:25:14 to 02:54:36. All four exited 0; job state succeeded.
- **Cards and receipts.** Launch cards X1-1, X2-1 and X4-1 have receipts (status completed). X1-1 and X4-1 show the wait-short step with exit 41, which I read as the job still running at the end of the 900 s wait (inferred from the job times above); the collect cards X1-V01, X3-V01, X4-V01 and X2-C01 (receipts, all steps exit 0, including the S3 hash check and upload-raw) took the results. **Card X3-1 (launch of the cap-6 draw-0 job) has no receipt:** a container restart killed it during its wait step after the launch was recorded. The job itself is recorded by receipts/X3/launch/result.json (launcher 74aeef5a..., pod b6596acd... Running at launch time) and by the collect card X3-V01.

2. The four rows (measured)

Source: results/X{1..4}/<pod>/runs/X-m2-t12-k<draw>-c<cap>.jsonl. grid_zeros from the .gridzeros.jsonl beside each.

row	draw	cap	degree_reached	inconsistent	basis_len	timed_out	oversize	max_cols	muladds	wall_s	grid_zeros	refutes
X1	0	5	5	false	2922	false	false	16572	179052499158	10.8	0	no
X2	1	5	5	false	2922	false	false	16572	179052499158	11.8	0	no
X3	0	6	6	true	1	false	false	103785	42213325264923	1824.0	0	yes
X4	1	6	6	true	1	false	false	103785	42213325353310	1762.8	0	yes

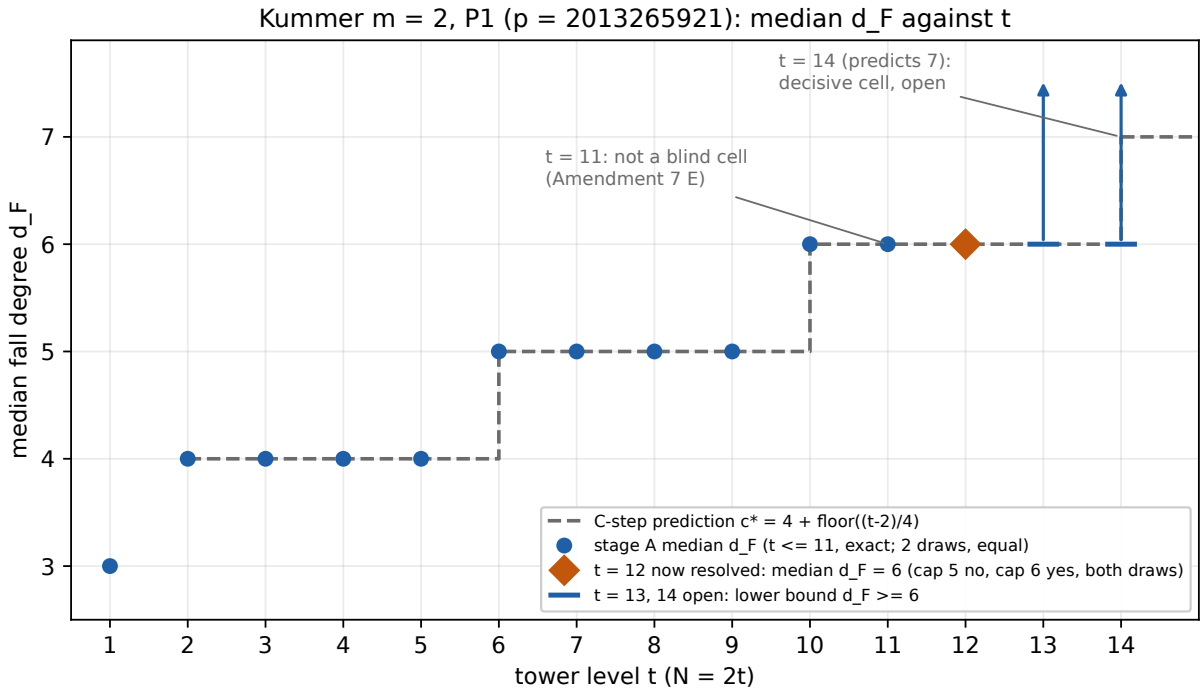
- Each row has zero_dim true, full_rank_exit true, planted_ok null (random target), and its x_r equals the one in the matching .gridzeros.jsonl row (150938225 for draw 0, 1002906059 for draw 1).
- **Repeat of stage A (derived).** Rows X1 and X2 equal stage A's results/R9/*/runs/CAP-m2-t12-k{0,1}-c5.jsonl in x_r, basis_len, max_cols, muladds, degree_reached and inconsistent: a repeat of an exact computation. Wall time differs (stage A 17.3 s and 17.1 s on another pod, with --budget 1800 --max-nnz 2e9; here 10.8 s and 11.8 s). For RESULTS.md U5 (whether the cap-5 step had to be repeated under the Amendment 7 A limits; the director's call), the repeat gives the same value.
- muladds is the count of F_p multiply-adds; it is deterministic and differs between the two cap-6 draws only in the 9th digit.

3. The bracket and the D.5 line (derived, by hand)

Amendment 7 B at t = 12: c* = 4 + floor((12 - 2)/4) = 6. Run cap c* - 1 = 5; if it does not refute, run cap c* = 6; if that refutes, d_F = c* = 6.

draw	cap 5	cap 6	d_F
0	does not refute (X1)	refutes (X3)	6
1	does not refute (X2)	refutes (X4)	6

- Median `d_F` = 6, exact. No row is censored: no `timed_out`, no `oversize`.
- Amendment 7 D.5: "holds at t (median equals c*)". **Reading: holds at t = 12.** Stage A's table 5 row for `t = 12` ("undecided", bracket started) is now resolved by these rows. The change is in the `t = 12` row only. `t = 11` (6, `=D (F4)`), `t = 13` and `t = 14` are unchanged.
- The closure bound of stage A (`d_F > 5`) agrees with the cap-5 step.



Sources: stage A pfd-a-20261108 (t <= 11), X1-X4 (t = 12), results/X12/analysis/pfdX12_summary.json. n = 2 draws per t, none decomposable. [pcc:unverified]

Figure 1: Median d_F against t, Kummer m = 2, P1. Sources: pfd-a-20261108 (t <= 11), jobs X1-X4 (t = 12); n = 2 draws per t. [pcc:unverified]

4. What this shows and does not show

Shows (all `[pcc:unverified]`): on two random draws at `t = 12`, capped F4 at degree 5 does not reach 1 and capped F4 at degree 6 does. Under the Amendment 7 B and D.5 rule text, that reads "holds at t = 12". The reading is consistent with the registered C-step value 6.

Does not show, each stated once:

- **Scorer.** The D.5 reading was applied from the rule text by hand and by a scratchpad script. The Amendment 7 D scorer patch is still not committed or reviewed (RESULTS.md section 8). The committed `prime_fall_degree_tools score` was not run on these rows.
- **Not a test that separates the readings.** `t = 12` predicts 6, the value already seen at `t = 11`. This cell cannot tell "C-step" from "flat at 6". Amendment 7 E names `t = 14` (predicting 7) as the decisive cell. `t = 13` and `t = 14` remain undecided (median >= 6; stage A's closure bound, no F4 bracket row).
- **Sample.** Two draws only, both not decomposable (`grid_zeros = 0`). A median of two equal values has no spread.
- **G3.** G3 (capped F4 agrees with the closure where both run) passed in `pfd-a` (RESULTS.md section 1: 144 capped runs on 72 closure systems with `N <= 20`, 0 disagreements, caps `d - 1` and `d` only; U3 records that this is a subset of what the text implies) and in the stage C G3-only job `pfd-g3c-20261109` (RESULTS_STAGE_C.md section 2a: 98 capped runs, 0 disagreements, on the `m <= 6` cells). These runs are `m = 2`, inside the F4 engine's capacity (`m <= 6`, Amendment 11). G3's tool tests `N <= 20`; `N = 24` here is outside that range, so no G3 run covers this cell itself.
- **Scope of the measurement.** This is a fall-degree measurement on toy primes (`P1 = 2013265921`), `m = 2` only, `t <= 14`. It is not an index-calculus cost result: no leaderboard row, no rho comparison, no `S`, and no improvement over the index-calculus baseline is claimed. `[pcc:unverified]`

5. Limits

- Instance type not recorded (section 1). Wall seconds depend on the host and are context only; no timing claim is made. No L2, replay or A/A admission applies to these command jobs (README).
- X1 and X2 used different request protocol hashes (section 1); the cap-5 rows are equal in every compared field, so the choice does not show in the data.
- Card `X3-1` has no receipt (section 1). The job's data and its collect receipt are intact.
- No pcc claim was filed. Every number is an unverified assertion until a claim and a signed verdict exist.

6. Evidence paths

Under `research/pkm_falldegree_unblock_20261107/`:

- Rows: `results/X1/9f2460de-3e12-455b-b34b-f1bfd5c4f9c4/`, `results/X2/16cf635f-d0be-45ce-afb1-244fe5b0b838/`, `results/X3/b6596acd-b2ef-49fd-befc-b5929de42242/`, `results/X4/6cb84cac-2967-42f7-b858-a1965d3c9892/` (each: `runs/`, `host.txt`, `progress.txt`, `binary.sha256`). These directories are git-excluded; the raw files are in S3 by SHA-256 through the record below.
- Requests: `requests_r2f/pfd-x-t12-k0-c5.json`, `requests_r2g/pfd-x-t12-k{1-c5,0-c6,1-c6}.json`.
- Receipts: `receipts/X1..X4/` (`launch/`, `status/`, `collect-*`); `cards/CARD-pkm-falldegree-unblock-20261107-pfd-x-t12-*.receipt.json`.
- Stage A rows compared: `results/R9/*/runs/CAP-m2-t12-k{0,1}-c5.jsonl`; stage A values for `t <= 11`: `results/R9/analysis/pfdA_summary.json`.
- Benchmark record: `benchmarks/records/20261109T2346Z-pkm-fd-pfd-x-t12.json` (commands, raw entries, metrics).
- This analysis: `results/X12/analysis/` holds `pfdX12_fig1.svg` and `.png` (median `d_F` against `t`, with the C-step step function; `t = 13, 14` drawn as open lower bounds), `pfdX12_report.pdf`, `pfdX12_summary.json` (the four rows and the reading) and `scripts-archive.json`. The generator scripts are kept out of the repository, as in stage A and C, in a tarball recorded in the benchmark record and stored in S3 by SHA-256 (`scripts-archive.json`).
- Rule text: `research/prime_fall_degree_20261027/PREREGISTRATION.md`, Amendment 6, Amendment 7 B, C, D and E, Amendment 11.